

SPPH 604 — In-class lecture slide plans (Weeks 1–7)

Slide-by-slide **blueprints** (not finished decks) for the weekly class session. Weeks 2–7 run **Tuesday 9:30 am–12 pm (150 minutes)**, ahead of that week’s Thursday lab (10 am–12 pm), so the lecture and lab feel continuous. **Week 1 is the exception:** there is no Tuesday session and no lab — the course opens with a **two-hour Thursday session** (Sept 10, 10 am–12 pm, by Zoom). Each lecture teaches the concept and theoretical issue of the demonstration paper (**MASLD / NHANES**, Kueh et al., BMJ Open 2026) *before* students go into that week’s lab.

Every plan has the same shape:

- **Learning objectives** (3–5)
- **Continuity** — recap of last week’s lab + what today sets up
- a numbered **slide blueprint** — each slide has content bullets, a “**MASLD hook**” tying it to the paper, and an approximate minute count (with a break marker)
- a closing “**Bridge to today’s lab**” slide stating exactly what students do in Lx and how it becomes their increment Px

The seven weeks

Week	Lecture	Feeds lab	Feeds increment
1	Course intro + procedures (+ 1 paper teaser)	— (setup)	begins M0 shortlisting
2	NHANES data sources; reading Methods critically	L1 Data wrangling	M0 shortlist
3	Cohort construction — FLI, MASLD criteria, eligibility	L2 Analytic data	P1 eligibility (M0 locked)
4	Confounding & DAGs; Table 1; variable roles	L3 Confounding	P2 Table 1
5	Interaction & effect modification	L4 Interaction	P3 effect modification
6	Complex survey design (weights/strata/PSU)	L5 Survey analysis	P4 design-aware estimate
7	Missing data (MCAR/MAR/MNAR; MI)	L6 Missing data	P5 missing-data sensitivity

Design notes

- **Continuity is the point.** Slide 1 of each week recaps where the project stands; the last slide hands off directly into the lab and the formative increment.
- **The paper is the running example.** Concepts are taught through the MASLD reproduction: Group IV HR $\approx 15 \rightarrow 2.9$, the sex interaction that is null unweighted and moves under the design, FLI circularity, the survey-weighting attenuation, the typed cohort funnel. The cohort size is the first lesson rather than a verified match — the paper reports $N \approx 6,300$, our full analytic file is 6,371, and accounting for the difference is the skill being taught.
- **The weeks build toward M1.** The demonstration paper runs through Weeks 3–5 as sustained practice in replication and methodological interrogation, so the analytic judgment **M1** asks each group to make rests on skills they have already rehearsed — the judgment itself stays theirs.

- **Timing convention — budget to the clock.** The session length is the cap: **150 min** for a Tuesday class (Weeks 2–7), **120 min** for the Week 1 Thursday opener. A plan’s own minute markers are the whole budget and the only accounting: they include the break, and nothing outside them is counted, because the untimed **Learning objectives** and **Continuity** blocks at the head of every plan are covered inside that week’s Slide 1. Markers may sum *to* the cap; they may never sum *past* it. Whatever is left over is **float** — minutes the room spends on questions and the vote-and-argue moments. Buffer is carried *inside* the plan, never left as unallocated time at the end: each plan names one recap or preview slide as its **designated cut**, so an overrun costs a slide chosen in advance instead of the bridge into the lab. The plan file is the budget of record; the decks’ speaker-note timings re-allocate a plan’s minutes and are not summed here.

Checking the timings

Each plan states its own total on its **Duration:** line. That line is the claim; this is the check, and it re-derives all seven weeks at once whenever a plan is retimed (Git Bash, from the repo root):

```
python -c "import re,glob;P=re.compile(r'(?m)^(?:\#\#\s.*?(~?(\d+) ?min\\)|\s*[-*]\s*(~?(\d+) ?min\\)?
```

Read each total against that week’s cap — 120 for Week 1, 150 for Weeks 2–7. Only what the command matches counts as a marker: minutes written into a plan’s prose are not part of the budget, so a total counted by eye can come out higher than the real one. A week over its cap owes a trim in its own file, not a footnote here. This README publishes no table of the current totals on purpose — it would be a frozen snapshot of seven files it does not own, and stale the first time one of them is retimed.

Built decks (slides/)

The plans have been turned into finished **Quarto revealjs decks** in `slides/` — one `.qmd` per week plus a self-contained rendered `.html` (open in any browser; arrow keys to navigate, S for speaker notes, F for fullscreen). Each deck has a title slide, objectives, a continuity slide, the plan’s content slides with **MASLD** running-example callouts, and detailed **speaker notes** (timings + teaching prompts) under each slide. The Kaplan–Meier figure is embedded on the weeks that discuss the survival result (2, 5, 6). Each `.html` is the rendered form of the `.qmd` beside it, so present from the `.html` and treat the `.qmd` as the source. If you adapt a deck, change the `.qmd` and re-render it with Quarto rather than editing the `.html`. From this directory:

```
quarto render slides/Week5_interaction_slides.qmd
```

`img/` holds the two embedded figures, copied from `../reproduction/output/figures/`.